

Filter Summary Report: CG,TIA,Automated,NO,L,simple,Z1,Z3,Z4

Generated by MacAnalog-Symbolix

October 17, 2025

Contents

1 Examined $H(z)$ for CG TIA Automated NO L simple Z1 Z3 Z4: $\frac{Z_1 Z_3 Z_4 g_m}{2 Z_1 Z_3 g_m + Z_1 Z_4 g_m + 2 Z_3 + Z_4}$

$$H(z) = \frac{Z_1 Z_3 Z_4 g_m}{2 Z_1 Z_3 g_m + Z_1 Z_4 g_m + 2 Z_3 + Z_4}$$

2 AP

3 BP

4 BS

5 GE

6 HP

7 LP

7.1 LP-1 $Z(s) = \left(\frac{1}{C_1 s}, \infty, R_3, \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{R_3 g_m}{2 C_1 C_4 R_3 s^2 + g_m + s (C_1 + 2 C_4 R_3 g_m)}$$

Parameters:

Q: $\frac{\sqrt{2} \sqrt{C_1} \sqrt{C_4} \sqrt{R_3} \sqrt{g_m}}{C_1 + 2 C_4 R_3 g_m}$
wo: $\frac{\sqrt{2} \sqrt{g_m}}{2 \sqrt{C_1} \sqrt{C_4} \sqrt{R_3}}$
bandwidth: $\frac{C_1 + 2 C_4 R_3 g_m}{2 C_1 C_4 R_3}$
K-LP: R_3
K-HP: 0
K-BP: 0
Qz: None
Wz: None

7.2 LP-2 $Z(s) = \left(\frac{1}{C_1 s}, \infty, R_3, \frac{R_4}{C_4 R_4 s + 1}, \infty, \infty \right)$

$$H(s) = \frac{R_3 R_4 g_m}{2 C_1 C_4 R_3 R_4 s^2 + 2 R_3 g_m + R_4 g_m + s (2 C_1 R_3 + C_1 R_4 + 2 C_4 R_3 R_4 g_m)}$$

Parameters:

Q: $\frac{\sqrt{2} \sqrt{C_1} \sqrt{C_4} \sqrt{R_3} \sqrt{R_4} \sqrt{g_m} \sqrt{2 R_3 + R_4}}{2 C_1 R_3 + C_1 R_4 + 2 C_4 R_3 R_4 g_m}$
wo: $\frac{\sqrt{2} \sqrt{2 R_3 g_m + R_4 g_m}}{2 \sqrt{C_1} \sqrt{C_4} \sqrt{R_3} \sqrt{R_4}}$
bandwidth: $\frac{\sqrt{2 R_3 g_m + R_4 g_m} (2 C_1 R_3 + C_1 R_4 + 2 C_4 R_3 R_4 g_m)}{2 C_1 C_4 R_3 R_4 \sqrt{g_m} \sqrt{2 R_3 + R_4}}$
K-LP: $\frac{R_3 R_4}{2 R_3 + R_4}$
K-HP: 0
K-BP: 0
Qz: None
Wz: None

7.3 LP-3 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \frac{1}{C_3 s}, R_4, \infty, \infty \right)$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_1}\sqrt{C_3}\sqrt{R_4}\sqrt{g_m}}{2C_1+C_3R_4g_m}$
 wo: $\frac{\sqrt{2}\sqrt{g_m}}{\sqrt{C_1}\sqrt{C_3}\sqrt{R_4}}$
 bandwidth: $\frac{2C_1+C_3R_4g_m}{C_1C_3R_4}$
 K-LP: $\frac{R_4}{2}$
 K-HP: 0
 K-BP: 0
 Qz: None
 Wz: None

$$H(s) = \frac{R_4g_m}{C_1C_3R_4s^2 + 2g_m + s(2C_1 + C_3R_4g_m)}$$

7.4 LP-4 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \frac{1}{C_3 s}, \frac{R_4}{C_4R_4s+1}, \infty, \infty \right)$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_1}\sqrt{R_4}\sqrt{g_m}\sqrt{C_3+2C_4}}{2C_1+C_3R_4g_m+2C_4R_4g_m}$
 wo: $\frac{\sqrt{2}\sqrt{g_m}}{\sqrt{C_1C_3R_4+2C_1C_4R_4}}$
 bandwidth: $\frac{2C_1+C_3R_4g_m+2C_4R_4g_m}{\sqrt{C_1}\sqrt{R_4}\sqrt{C_3+2C_4}\sqrt{C_1C_3R_4+2C_1C_4R_4}}$
 K-LP: $\frac{R_4}{2}$
 K-HP: 0
 K-BP: 0
 Qz: None
 Wz: None

$$H(s) = \frac{R_4g_m}{2g_m + s^2(C_1C_3R_4 + 2C_1C_4R_4) + s(2C_1 + C_3R_4g_m + 2C_4R_4g_m)}$$

7.5 LP-5 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \frac{R_3}{C_3R_3s+1}, R_4, \infty, \infty \right)$

Parameters:

Q: $\frac{\sqrt{C_1}\sqrt{C_3}\sqrt{R_3}\sqrt{R_4}\sqrt{g_m}\sqrt{2R_3+R_4}}{2C_1R_3+C_1R_4+C_3R_3R_4g_m}$
 wo: $\frac{\sqrt{2R_3g_m+R_4g_m}}{\sqrt{C_1}\sqrt{C_3}\sqrt{R_3}\sqrt{R_4}}$
 bandwidth: $\frac{\sqrt{2R_3g_m+R_4g_m}(2C_1R_3+C_1R_4+C_3R_3R_4g_m)}{C_1C_3R_3R_4\sqrt{g_m}\sqrt{2R_3+R_4}}$
 K-LP: $\frac{R_3R_4}{2R_3+R_4}$
 K-HP: 0
 K-BP: 0
 Qz: None
 Wz: None

$$H(s) = \frac{R_3R_4g_m}{C_1C_3R_3R_4s^2 + 2R_3g_m + R_4g_m + s(2C_1R_3 + C_1R_4 + C_3R_3R_4g_m)}$$

7.6 LP-6 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \frac{R_3}{C_3R_3s+1}, \frac{1}{C_4 s}, \infty, \infty \right)$

Parameters:

Q: $\frac{\sqrt{C_1}\sqrt{R_3}\sqrt{g_m}\sqrt{C_3+2C_4}}{C_1+C_3R_3g_m+2C_4R_3g_m}$
 wo: $\frac{\sqrt{g_m}}{\sqrt{C_1C_3R_3+2C_1C_4R_3}}$

$$H(s) = \frac{R_3g_m}{g_m + s^2(C_1C_3R_3 + 2C_1C_4R_3) + s(C_1 + C_3R_3g_m + 2C_4R_3g_m)}$$

bandwidth: $\frac{C_1+C_3R_3g_m+2C_4R_3g_m}{\sqrt{C_1}\sqrt{R_3}\sqrt{C_3+2C_4}\sqrt{C_1C_3R_3+2C_1C_4R_3}}$
K-LP: R_3
K-HP: 0
K-BP: 0
Qz: None
Wz: None

7.7 LP-7 $Z(s) = \left(\frac{1}{C_1s}, \infty, \frac{R_3}{C_3R_3s+1}, \frac{R_4}{C_4R_4s+1}, \infty, \infty \right)$

$$H(s) = \frac{R_3R_4g_m}{2R_3g_m + R_4g_m + s^2 (C_1C_3R_3R_4 + 2C_1C_4R_3R_4) + s (2C_1R_3 + C_1R_4 + C_3R_3R_4g_m + 2C_4R_3R_4g_m)}$$

Parameters:

Q: $\frac{\sqrt{C_1}\sqrt{R_3}\sqrt{R_4}\sqrt{g_m}\sqrt{C_3+2C_4}\sqrt{2R_3+R_4}}{2C_1R_3+C_1R_4+C_3R_3R_4g_m+2C_4R_3R_4g_m}$
wo: $\frac{\sqrt{2R_3g_m+R_4g_m}}{\sqrt{C_1C_3R_3R_4+2C_1C_4R_3R_4}}$
bandwidth: $\frac{\sqrt{2R_3g_m+R_4g_m}(2C_1R_3+C_1R_4+C_3R_3R_4g_m+2C_4R_3R_4g_m)}{\sqrt{C_1}\sqrt{R_3}\sqrt{R_4}\sqrt{g_m}\sqrt{C_3+2C_4}\sqrt{2R_3+R_4}\sqrt{C_1C_3R_3R_4+2C_1C_4R_3R_4}}$
K-LP: $\frac{R_3R_4}{2R_3+R_4}$
K-HP: 0
K-BP: 0
Qz: None
Wz: None

7.8 LP-8 $Z(s) = \left(\frac{R_1}{C_1R_1s+1}, \infty, R_3, \frac{1}{C_4s}, \infty, \infty \right)$

$$H(s) = \frac{R_1R_3g_m}{2C_1C_4R_1R_3s^2 + R_1g_m + s (C_1R_1 + 2C_4R_1R_3g_m + 2C_4R_3) + 1}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_1}\sqrt{C_4}\sqrt{R_1}\sqrt{R_3}\sqrt{R_1g_m+1}}{C_1R_1+2C_4R_1R_3g_m+2C_4R_3}$
wo: $\frac{\sqrt{2}\sqrt{R_1g_m+1}}{2\sqrt{C_1}\sqrt{C_4}\sqrt{R_1}\sqrt{R_3}}$
bandwidth: $\frac{C_1R_1+2C_4R_1R_3g_m+2C_4R_3}{2C_1C_4R_1R_3}$
K-LP: $\frac{R_1R_3g_m}{R_1g_m+1}$
K-HP: 0
K-BP: 0
Qz: None
Wz: None

7.9 LP-9 $Z(s) = \left(\frac{R_1}{C_1R_1s+1}, \infty, R_3, \frac{R_4}{C_4R_4s+1}, \infty, \infty \right)$

$$H(s) = \frac{R_1R_3R_4g_m}{2C_1C_4R_1R_3R_4s^2 + 2R_1R_3g_m + R_1R_4g_m + 2R_3 + R_4 + s (2C_1R_1R_3 + C_1R_1R_4 + 2C_4R_1R_3R_4g_m + 2C_4R_3R_4)}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_1}\sqrt{C_4}\sqrt{R_1}\sqrt{R_3}\sqrt{R_4}\sqrt{2R_1R_3g_m+R_1R_4g_m+2R_3+R_4}}{2C_1R_1R_3+C_1R_1R_4+2C_4R_1R_3R_4g_m+2C_4R_3R_4}$
wo: $\frac{\sqrt{2}\sqrt{2R_1R_3g_m+R_1R_4g_m+2R_3+R_4}}{2\sqrt{C_1}\sqrt{C_4}\sqrt{R_1}\sqrt{R_3}\sqrt{R_4}}$
bandwidth: $\frac{2C_1R_1R_3+C_1R_1R_4+2C_4R_1R_3R_4g_m+2C_4R_3R_4}{2C_1C_4R_1R_3R_4}$
K-LP: $\frac{R_1R_3R_4g_m}{2R_1R_3g_m+R_1R_4g_m+2R_3+R_4}$
K-HP: 0
K-BP: 0
Qz: None
Wz: None

7.10 LP-10 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \frac{1}{C_3 s}, R_4, \infty, \infty \right)$

$$H(s) = \frac{R_1 R_4 g_m}{C_1 C_3 R_1 R_4 s^2 + 2 R_1 g_m + s(2 C_1 R_1 + C_3 R_1 R_4 g_m + C_3 R_4) + 2}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_1}\sqrt{C_3}\sqrt{R_1}\sqrt{R_4}\sqrt{R_1 g_m + 1}}{2 C_1 R_1 + C_3 R_1 R_4 g_m + C_3 R_4}$
wo: $\frac{\sqrt{2 R_1 g_m + 2}}{\sqrt{C_1}\sqrt{C_3}\sqrt{R_1}\sqrt{R_4}}$
bandwidth: $\frac{\sqrt{2}\sqrt{2 R_1 g_m + 2}(2 C_1 R_1 + C_3 R_1 R_4 g_m + C_3 R_4)}{2 C_1 C_3 R_1 R_4 \sqrt{R_1 g_m + 1}}$
K-LP: $\frac{R_1 R_4 g_m}{2 R_1 g_m + 2}$
K-HP: 0
K-BP: 0
Qz: None
Wz: None

7.11 LP-11 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \frac{1}{C_3 s}, \frac{R_4}{C_4 R_4 s + 1}, \infty, \infty \right)$

$$H(s) = \frac{R_1 R_4 g_m}{2 R_1 g_m + s^2(C_1 C_3 R_1 R_4 + 2 C_1 C_4 R_1 R_4) + s(2 C_1 R_1 + C_3 R_1 R_4 g_m + C_3 R_4 + 2 C_4 R_1 R_4 g_m + 2 C_4 R_4) + 2}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_1}\sqrt{R_1}\sqrt{R_4}\sqrt{C_3 + 2 C_4}\sqrt{R_1 g_m + 1}}{2 C_1 R_1 + C_3 R_1 R_4 g_m + C_3 R_4 + 2 C_4 R_1 R_4 g_m + 2 C_4 R_4}$
wo: $\frac{\sqrt{2 R_1 g_m + 2}}{\sqrt{C_1 C_3 R_1 R_4 + 2 C_1 C_4 R_1 R_4}}$
bandwidth: $\frac{\sqrt{2}\sqrt{2 R_1 g_m + 2}(2 C_1 R_1 + C_3 R_1 R_4 g_m + C_3 R_4 + 2 C_4 R_1 R_4 g_m + 2 C_4 R_4)}{2\sqrt{C_1}\sqrt{R_1}\sqrt{R_4}\sqrt{C_3 + 2 C_4}\sqrt{R_1 g_m + 1}\sqrt{C_1 C_3 R_1 R_4 + 2 C_1 C_4 R_1 R_4}}$
K-LP: $\frac{R_1 R_4 g_m}{2 R_1 g_m + 2}$
K-HP: 0
K-BP: 0
Qz: None
Wz: None

7.12 LP-12 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \frac{R_3}{C_3 R_3 s + 1}, R_4, \infty, \infty \right)$

$$H(s) = \frac{R_1 R_3 R_4 g_m}{C_1 C_3 R_1 R_3 R_4 s^2 + 2 R_1 R_3 g_m + R_1 R_4 g_m + 2 R_3 + R_4 + s(2 C_1 R_1 R_3 + C_1 R_1 R_4 + C_3 R_1 R_3 R_4 g_m + C_3 R_3 R_4)}$$

Parameters:

Q: $\frac{\sqrt{C_1}\sqrt{C_3}\sqrt{R_1}\sqrt{R_3}\sqrt{R_4}\sqrt{2 R_1 R_3 g_m + R_1 R_4 g_m + 2 R_3 + R_4}}{2 C_1 R_1 R_3 + C_1 R_1 R_4 + C_3 R_1 R_3 R_4 g_m + C_3 R_3 R_4}$
wo: $\frac{\sqrt{2 R_1 R_3 g_m + R_1 R_4 g_m + 2 R_3 + R_4}}{\sqrt{C_1}\sqrt{C_3}\sqrt{R_1}\sqrt{R_3}\sqrt{R_4}}$
bandwidth: $\frac{2 C_1 R_1 R_3 + C_1 R_1 R_4 + C_3 R_1 R_3 R_4 g_m + C_3 R_3 R_4}{C_1 C_3 R_1 R_3 R_4}$
K-LP: $\frac{R_1 R_3 R_4 g_m}{2 R_1 R_3 g_m + R_1 R_4 g_m + 2 R_3 + R_4}$
K-HP: 0
K-BP: 0
Qz: None
Wz: None

7.13 LP-13 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \frac{R_3}{C_3 R_3 s + 1}, \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{R_1 R_3 g_m}{R_1 g_m + s^2(C_1 C_3 R_1 R_3 + 2 C_1 C_4 R_1 R_3) + s(C_1 R_1 + C_3 R_1 R_3 g_m + C_3 R_3 + 2 C_4 R_1 R_3 g_m + 2 C_4 R_3) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_1}\sqrt{R_1}\sqrt{R_3}\sqrt{C_3 + 2 C_4}\sqrt{R_1 g_m + 1}}{C_1 R_1 + C_3 R_1 R_3 g_m + C_3 R_3 + 2 C_4 R_1 R_3 g_m + 2 C_4 R_3}$
wo: $\frac{\sqrt{R_1 g_m + 1}}{\sqrt{C_1 C_3 R_1 R_3 + 2 C_1 C_4 R_1 R_3}}$

bandwidth: $\frac{C_1 R_1 + C_3 R_1 R_3 g_m + C_3 R_3 + 2C_4 R_1 R_3 g_m + 2C_4 R_3}{\sqrt{C_1} \sqrt{R_1} \sqrt{R_3} \sqrt{C_3 + 2C_4} \sqrt{C_1 C_3 R_1 R_3 + 2C_1 C_4 R_1 R_3}}$
K-LP: $\frac{R_1 R_3 g_m}{R_1 g_m + 1}$
K-HP: 0
K-BP: 0
Qz: None
Wz: None

7.14 LP-14 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \frac{R_3}{C_3 R_3 s + 1}, \frac{R_4}{C_4 R_4 s + 1}, \infty, \infty \right)$

$$H(s) = \frac{R_1 R_3 R_4 g_m}{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4 + s^2 (C_1 C_3 R_1 R_3 R_4 + 2C_1 C_4 R_1 R_3 R_4) + s (2C_1 R_1 R_3 + C_1 R_1 R_4 + C_3 R_1 R_3 R_4 g_m + C_3 R_3 R_4 + 2C_4 R_1 R_3 R_4 g_m + 2C_4 R_3 R_4)}$$

Parameters:

Q: $\frac{\sqrt{C_1} \sqrt{R_1} \sqrt{R_3} \sqrt{R_4} \sqrt{C_3 + 2C_4} \sqrt{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4}}{2C_1 R_1 R_3 + C_1 R_1 R_4 + C_3 R_1 R_3 R_4 g_m + C_3 R_3 R_4 + 2C_4 R_1 R_3 R_4 g_m + 2C_4 R_3 R_4}$
wo: $\frac{\sqrt{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4}}{\sqrt{C_1} C_3 R_1 R_3 R_4 + 2C_1 C_4 R_1 R_3 R_4}$
bandwidth: $\frac{2C_1 R_1 R_3 + C_1 R_1 R_4 + C_3 R_1 R_3 R_4 g_m + C_3 R_3 R_4 + 2C_4 R_1 R_3 R_4 g_m + 2C_4 R_3 R_4}{\sqrt{C_1} \sqrt{R_1} \sqrt{R_3} \sqrt{R_4} \sqrt{C_3 + 2C_4} \sqrt{C_1 C_3 R_1 R_3 R_4 + 2C_1 C_4 R_1 R_3 R_4}}$
K-LP: $\frac{R_1 R_3 R_4 g_m}{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4}$
K-HP: 0
K-BP: 0
Qz: None
Wz: None

8 X-INVALID-NUMER

8.1 X-INVALID-NUMER-1 $Z(s) = \left(R_1, \infty, \frac{R_3}{C_3 R_3 s + 1}, R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_4 R_1 R_3 R_4 g_m s + R_1 R_3 g_m}{R_1 g_m + s^2 (C_3 C_4 R_1 R_3 R_4 g_m + C_3 C_4 R_3 R_4) + s (C_3 R_1 R_3 g_m + C_3 R_3 + 2C_4 R_1 R_3 g_m + C_4 R_1 R_4 g_m + 2C_4 R_3 + C_4 R_4) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_3} \sqrt{C_4} \sqrt{R_3} \sqrt{R_4}}{C_3 R_3 + 2C_4 R_3 + C_4 R_4}$
wo: $\frac{1}{\sqrt{C_3} \sqrt{C_4} \sqrt{R_3} \sqrt{R_4}}$
bandwidth: $\frac{C_3 R_3 + 2C_4 R_3 + C_4 R_4}{C_3 C_4 R_3 R_4}$
K-LP: $\frac{R_1 R_3 g_m}{R_1 g_m + 1}$
K-HP: 0
K-BP: $\frac{C_4 R_1 R_3 R_4 g_m}{C_3 R_1 R_3 g_m + C_3 R_3 + 2C_4 R_1 R_3 g_m + C_4 R_1 R_4 g_m + 2C_4 R_3 + C_4 R_4}$
Qz: None
Wz: None

8.2 X-INVALID-NUMER-2 $Z(s) = \left(R_1, \infty, R_3 + \frac{1}{C_3 s}, \frac{R_4}{C_4 R_4 s + 1}, \infty, \infty \right)$

$$H(s) = \frac{C_3 R_1 R_3 R_4 g_m s + R_1 R_4 g_m}{2R_1 g_m + s^2 (2C_3 C_4 R_1 R_3 R_4 g_m + 2C_3 C_4 R_3 R_4) + s (2C_3 R_1 R_3 g_m + C_3 R_1 R_4 g_m + 2C_3 R_3 + C_3 R_4 + 2C_4 R_1 R_4 g_m + 2C_4 R_4) + 2}$$

Parameters:

Q: $\frac{2\sqrt{C_3} \sqrt{C_4} \sqrt{R_3} \sqrt{R_4}}{2C_3 R_3 + C_3 R_4 + 2C_4 R_4}$
wo: $\frac{1}{\sqrt{C_3} \sqrt{C_4} \sqrt{R_3} \sqrt{R_4}}$
bandwidth: $\frac{2C_3 R_3 + C_3 R_4 + 2C_4 R_4}{2C_3 C_4 R_3 R_4}$
K-LP: $\frac{R_1 R_4 g_m}{2R_1 g_m + 2}$
K-HP: 0
K-BP: $\frac{C_3 R_1 R_3 R_4 g_m}{2C_3 R_1 R_3 g_m + C_3 R_1 R_4 g_m + 2C_3 R_3 + C_3 R_4 + 2C_4 R_1 R_4 g_m + 2C_4 R_4}$
Qz: None
Wz: None

8.3 X-INVALID-NUMER-3 $Z(s) = \left(\frac{1}{C_1 s}, \infty, R_3, R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_4 R_3 R_4 g_m s + R_3 g_m}{g_m + s^2 (2C_1 C_4 R_3 + C_1 C_4 R_4) + s (C_1 + 2C_4 R_3 g_m + C_4 R_4 g_m)}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{C_1} \sqrt{C_4} \sqrt{g_m} \sqrt{2R_3 + R_4}}{C_1 + 2C_4 R_3 g_m + C_4 R_4 g_m} \\ \text{wo: } & \frac{\sqrt{g_m}}{\sqrt{2C_1 C_4 R_3 + C_1 C_4 R_4}} \\ \text{bandwidth: } & \frac{C_1 + 2C_4 R_3 g_m + C_4 R_4 g_m}{\sqrt{C_1} \sqrt{C_4} \sqrt{2R_3 + R_4} \sqrt{2C_1 C_4 R_3 + C_1 C_4 R_4}} \\ \text{K-LP: } & R_3 \\ \text{K-HP: } & 0 \\ \text{K-BP: } & \frac{C_4 R_3 R_4 g_m}{C_1 + 2C_4 R_3 g_m + C_4 R_4 g_m} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \text{None} \end{aligned}$$

8.4 X-INVALID-NUMER-4 $Z(s) = \left(\frac{1}{C_1 s}, \infty, R_3 + \frac{1}{C_3 s}, R_4, \infty, \infty \right)$

$$H(s) = \frac{C_3 R_3 R_4 g_m s + R_4 g_m}{2g_m + s^2 (2C_1 C_3 R_3 + C_1 C_3 R_4) + s (2C_1 + 2C_3 R_3 g_m + C_3 R_4 g_m)}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{2} \sqrt{C_1} \sqrt{C_3} \sqrt{g_m} \sqrt{2R_3 + R_4}}{2C_1 + 2C_3 R_3 g_m + C_3 R_4 g_m} \\ \text{wo: } & \frac{\sqrt{2} \sqrt{g_m}}{\sqrt{2C_1 C_3 R_3 + C_1 C_3 R_4}} \\ \text{bandwidth: } & \frac{2C_1 + 2C_3 R_3 g_m + C_3 R_4 g_m}{\sqrt{C_1} \sqrt{C_3} \sqrt{2R_3 + R_4} \sqrt{2C_1 C_3 R_3 + C_1 C_3 R_4}} \\ \text{K-LP: } & \frac{R_4}{2} \\ \text{K-HP: } & 0 \\ \text{K-BP: } & \frac{C_3 R_3 R_4 g_m}{2C_1 + 2C_3 R_3 g_m + C_3 R_4 g_m} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \text{None} \end{aligned}$$

8.5 X-INVALID-NUMER-5 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, R_3, R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_4 R_1 R_3 R_4 g_m s + R_1 R_3 g_m}{R_1 g_m + s^2 (2C_1 C_4 R_1 R_3 + C_1 C_4 R_1 R_4) + s (C_1 R_1 + 2C_4 R_1 R_3 g_m + C_4 R_1 R_4 g_m + 2C_4 R_3 + C_4 R_4) + 1}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{C_1} \sqrt{C_4} \sqrt{R_1} \sqrt{2R_3 + R_4} \sqrt{R_1 g_m + 1}}{C_1 R_1 + 2C_4 R_1 R_3 g_m + C_4 R_1 R_4 g_m + 2C_4 R_3 + C_4 R_4} \\ \text{wo: } & \frac{\sqrt{R_1 g_m + 1}}{\sqrt{2C_1 C_4 R_1 R_3 + C_1 C_4 R_1 R_4}} \\ \text{bandwidth: } & \frac{C_1 R_1 + 2C_4 R_1 R_3 g_m + C_4 R_1 R_4 g_m + 2C_4 R_3 + C_4 R_4}{\sqrt{C_1} \sqrt{C_4} \sqrt{R_1} \sqrt{2R_3 + R_4} \sqrt{2C_1 C_4 R_1 R_3 + C_1 C_4 R_1 R_4}} \\ \text{K-LP: } & \frac{R_1 R_3 g_m}{R_1 g_m + 1} \\ \text{K-HP: } & 0 \\ \text{K-BP: } & \frac{C_4 R_1 R_3 R_4 g_m}{C_1 R_1 + 2C_4 R_1 R_3 g_m + C_4 R_1 R_4 g_m + 2C_4 R_3 + C_4 R_4} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \text{None} \end{aligned}$$

8.6 X-INVALID-NUMER-6 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, R_3 + \frac{1}{C_3 s}, R_4, \infty, \infty \right)$

$$H(s) = \frac{C_3 R_1 R_3 R_4 g_m s + R_1 R_4 g_m}{2R_1 g_m + s^2 (2C_1 C_3 R_1 R_3 + C_1 C_3 R_1 R_4) + s (2C_1 R_1 + 2C_3 R_1 R_3 g_m + C_3 R_1 R_4 g_m + 2C_3 R_3 + C_3 R_4) + 2}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{2} \sqrt{C_1} \sqrt{C_3} \sqrt{R_1} \sqrt{2R_3 + R_4} \sqrt{R_1 g_m + 1}}{2C_1 R_1 + 2C_3 R_1 R_3 g_m + C_3 R_1 R_4 g_m + 2C_3 R_3 + C_3 R_4} \\ \text{wo: } & \frac{\sqrt{2R_1 g_m + 2}}{\sqrt{2C_1 C_3 R_1 R_3 + C_1 C_3 R_1 R_4}} \\ \text{bandwidth: } & \frac{\sqrt{2} \sqrt{2R_1 g_m + 2} (2C_1 R_1 + 2C_3 R_1 R_3 g_m + C_3 R_1 R_4 g_m + 2C_3 R_3 + C_3 R_4)}{2\sqrt{C_1} \sqrt{C_3} \sqrt{R_1} \sqrt{2R_3 + R_4} \sqrt{R_1 g_m + 1} \sqrt{2C_1 C_3 R_1 R_3 + C_1 C_3 R_1 R_4}} \end{aligned}$$

K-LP: $\frac{R_1 R_4 g_m}{2 R_1 g_m + 2}$
K-HP: 0
K-BP: $\frac{C_3 R_1 R_3 R_4 g_m}{2 C_1 R_1 + 2 C_3 R_1 R_3 g_m + C_3 R_1 R_4 g_m + 2 C_3 R_3 + C_3 R_4}$
Qz: None
Wz: None

8.7 X-INVALID-NUMER-7 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, R_3, \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_1 R_1 R_3 g_m s + R_3 g_m}{g_m + s^2 (2 C_1 C_4 R_1 R_3 g_m + 2 C_1 C_4 R_3) + s (C_1 R_1 g_m + C_1 + 2 C_4 R_3 g_m)}$$

Parameters:

Q: $\frac{\sqrt{2} \sqrt{C_1} \sqrt{C_4} R_1 \sqrt{R_3 g_m} \sqrt{\frac{g_m}{R_1 g_m + 1}} + \sqrt{2} \sqrt{C_1} \sqrt{C_4} \sqrt{R_3} \sqrt{\frac{g_m}{R_1 g_m + 1}}}{C_1 R_1 g_m + C_1 + 2 C_4 R_3 g_m}$
wo: $\sqrt{\frac{g_m}{2 C_1 C_4 R_1 R_3 g_m + 2 C_1 C_4 R_3}}$
bandwidth: $\frac{\sqrt{\frac{g_m}{2 C_1 C_4 R_1 R_3 g_m + 2 C_1 C_4 R_3}} (C_1 R_1 g_m + C_1 + 2 C_4 R_3 g_m)}{\sqrt{2} \sqrt{C_1} \sqrt{C_4} R_1 \sqrt{R_3 g_m} \sqrt{\frac{g_m}{R_1 g_m + 1}} + \sqrt{2} \sqrt{C_1} \sqrt{C_4} \sqrt{R_3} \sqrt{\frac{g_m}{R_1 g_m + 1}}}$
K-LP: R_3
K-HP: 0
K-BP: $\frac{C_1 R_1 R_3 g_m}{C_1 R_1 g_m + C_1 + 2 C_4 R_3 g_m}$
Qz: None
Wz: None

8.8 X-INVALID-NUMER-8 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, R_3, \frac{R_4}{C_4 R_4 s + 1}, \infty, \infty \right)$

$$H(s) = \frac{C_1 R_1 R_3 R_4 g_m s + R_3 R_4 g_m}{2 R_3 g_m + R_4 g_m + s^2 (2 C_1 C_4 R_1 R_3 R_4 g_m + 2 C_1 C_4 R_3 R_4) + s (2 C_1 R_1 R_3 g_m + C_1 R_1 R_4 g_m + 2 C_1 R_3 + C_1 R_4 + 2 C_4 R_3 R_4 g_m)}$$

Parameters:

Q: $\frac{\sqrt{2} \sqrt{C_1} \sqrt{C_4} R_1 \sqrt{R_3} \sqrt{R_4 g_m} \sqrt{\frac{g_m}{R_1 g_m + 1}} \sqrt{2 R_3 + R_4} + \sqrt{2} \sqrt{C_1} \sqrt{C_4} \sqrt{R_3} \sqrt{R_4} \sqrt{\frac{g_m}{R_1 g_m + 1}} \sqrt{2 R_3 + R_4}}{2 C_1 R_1 R_3 g_m + C_1 R_1 R_4 g_m + 2 C_1 R_3 + C_1 R_4 + 2 C_4 R_3 R_4 g_m}$
wo: $\sqrt{\frac{2 R_3 g_m + R_4 g_m}{2 C_1 C_4 R_1 R_3 R_4 g_m + 2 C_1 C_4 R_3 R_4}}$
bandwidth: $\frac{\sqrt{\frac{2 R_3 g_m + R_4 g_m}{2 C_1 C_4 R_1 R_3 R_4 g_m + 2 C_1 C_4 R_3 R_4}} (2 C_1 R_1 R_3 g_m + C_1 R_1 R_4 g_m + 2 C_1 R_3 + C_1 R_4 + 2 C_4 R_3 R_4 g_m)}{\sqrt{2} \sqrt{C_1} \sqrt{C_4} R_1 \sqrt{R_3} \sqrt{R_4 g_m} \sqrt{\frac{g_m}{R_1 g_m + 1}} \sqrt{2 R_3 + R_4} + \sqrt{2} \sqrt{C_1} \sqrt{C_4} \sqrt{R_3} \sqrt{R_4} \sqrt{\frac{g_m}{R_1 g_m + 1}} \sqrt{2 R_3 + R_4}}$
K-LP: $\frac{R_3 R_4}{2 R_3 + R_4}$
K-HP: 0
K-BP: $\frac{C_1 R_1 R_3 R_4 g_m}{2 C_1 R_1 R_3 g_m + C_1 R_1 R_4 g_m + 2 C_1 R_3 + C_1 R_4 + 2 C_4 R_3 R_4 g_m}$
Qz: None
Wz: None

8.9 X-INVALID-NUMER-9 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \frac{1}{C_3 s}, R_4, \infty, \infty \right)$

$$H(s) = \frac{C_1 R_1 R_4 g_m s + R_4 g_m}{2 g_m + s^2 (C_1 C_3 R_1 R_4 g_m + C_1 C_3 R_4) + s (2 C_1 R_1 g_m + 2 C_1 + C_3 R_4 g_m)}$$

Parameters:

Q: $\frac{\sqrt{2} \sqrt{C_1} \sqrt{C_3} R_1 \sqrt{R_4 g_m} \sqrt{\frac{g_m}{R_1 g_m + 1}} + \sqrt{2} \sqrt{C_1} \sqrt{C_3} \sqrt{R_4} \sqrt{\frac{g_m}{R_1 g_m + 1}}}{2 C_1 R_1 g_m + 2 C_1 + C_3 R_4 g_m}$
wo: $\sqrt{2} \sqrt{\frac{g_m}{C_1 C_3 R_1 R_4 g_m + C_1 C_3 R_4}}$
bandwidth: $\frac{\sqrt{2} \sqrt{\frac{g_m}{C_1 C_3 R_1 R_4 g_m + C_1 C_3 R_4}} (2 C_1 R_1 g_m + 2 C_1 + C_3 R_4 g_m)}{\sqrt{2} \sqrt{C_1} \sqrt{C_3} R_1 \sqrt{R_4 g_m} \sqrt{\frac{g_m}{R_1 g_m + 1}} + \sqrt{2} \sqrt{C_1} \sqrt{C_3} \sqrt{R_4} \sqrt{\frac{g_m}{R_1 g_m + 1}}}$
K-LP: $\frac{R_4}{2}$
K-HP: 0
K-BP: $\frac{C_1 R_1 R_4 g_m}{2 C_1 R_1 g_m + 2 C_1 + C_3 R_4 g_m}$
Qz: None
Wz: None

8.10 X-INVALID-NUMER-10 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \frac{1}{C_3 s}, \frac{R_4}{C_4 R_4 s + 1}, \infty, \infty \right)$

$$H(s) = \frac{C_1 R_1 R_4 g_m s + R_4 g_m}{2g_m + s^2 (C_1 C_3 R_1 R_4 g_m + C_1 C_3 R_4 + 2C_1 C_4 R_1 R_4 g_m + 2C_1 C_4 R_4) + s (2C_1 R_1 g_m + 2C_1 + C_3 R_4 g_m + 2C_4 R_4 g_m)}$$

Parameters:

$$\text{Q: } \frac{\sqrt{2}\sqrt{C_1}C_3R_1\sqrt{R_4g_m}\sqrt{\frac{g_m}{C_3R_1g_m+C_3+2C_4R_1g_m+2C_4}}+\sqrt{2}\sqrt{C_1}C_3\sqrt{R_4}\sqrt{\frac{g_m}{C_3R_1g_m+C_3+2C_4R_1g_m+2C_4}}+2\sqrt{2}\sqrt{C_1}C_4R_1\sqrt{R_4g_m}\sqrt{\frac{g_m}{C_3R_1g_m+C_3+2C_4R_1g_m+2C_4}}+2\sqrt{2}\sqrt{C_1}C_4\sqrt{R_4}\sqrt{\frac{g_m}{C_3R_1g_m+C_3+2C_4R_1g_m+2C_4}}}{2C_1R_1g_m+2C_1+C_3R_4g_m+2C_4R_4g_m}$$

$$\text{wo: } \sqrt{2}\sqrt{\frac{g_m}{C_1C_3R_1R_4g_m+C_1C_3R_4+2C_1C_4R_1R_4g_m+2C_1C_4R_4}}$$

$$\text{bandwidth: } \frac{\sqrt{2}\sqrt{C_1}C_3R_1\sqrt{R_4g_m}\sqrt{\frac{g_m}{C_3R_1g_m+C_3+2C_4R_1g_m+2C_4}}+\sqrt{2}\sqrt{C_1}C_3\sqrt{R_4}\sqrt{\frac{g_m}{C_3R_1g_m+C_3+2C_4R_1g_m+2C_4}}+2\sqrt{2}\sqrt{C_1}C_4R_1\sqrt{R_4g_m}\sqrt{\frac{g_m}{C_3R_1g_m+C_3+2C_4R_1g_m+2C_4}}+2\sqrt{2}\sqrt{C_1}C_4\sqrt{R_4}\sqrt{\frac{g_m}{C_3R_1g_m+C_3+2C_4R_1g_m+2C_4}}}{\sqrt{2}\sqrt{C_1}C_3R_1\sqrt{R_4g_m}\sqrt{\frac{g_m}{C_3R_1g_m+C_3+2C_4R_1g_m+2C_4}}+\sqrt{2}\sqrt{C_1}C_3\sqrt{R_4}\sqrt{\frac{g_m}{C_3R_1g_m+C_3+2C_4R_1g_m+2C_4}}+2\sqrt{2}\sqrt{C_1}C_4R_1\sqrt{R_4g_m}\sqrt{\frac{g_m}{C_3R_1g_m+C_3+2C_4R_1g_m+2C_4}}+2\sqrt{2}\sqrt{C_1}C_4\sqrt{R_4}\sqrt{\frac{g_m}{C_3R_1g_m+C_3+2C_4R_1g_m+2C_4}}}$$

$$\text{K-LP: } \frac{R_4}{2}$$

$$\text{K-HP: } 0$$

$$\text{K-BP: } \frac{C_1R_1R_4g_m}{2C_1R_1g_m+2C_1+C_3R_4g_m+2C_4R_4g_m}$$

$$\text{Qz: None}$$

$$\text{Wz: None}$$

8.11 X-INVALID-NUMER-11 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \frac{R_3}{C_3 R_3 s + 1}, R_4, \infty, \infty \right)$

$$H(s) = \frac{C_1 R_1 R_3 R_4 g_m s + R_3 R_4 g_m}{2R_3g_m + R_4g_m + s^2 (C_1 C_3 R_1 R_3 R_4 g_m + C_1 C_3 R_3 R_4) + s (2C_1 R_1 R_3 g_m + C_1 R_1 R_4 g_m + 2C_1 R_3 + C_1 R_4 + C_3 R_3 R_4 g_m)}$$

Parameters:

$$\text{Q: } \frac{\sqrt{C_1}\sqrt{C_3}R_1\sqrt{R_3}\sqrt{R_4g_m}\sqrt{\frac{g_m}{R_1g_m+1}}\sqrt{2R_3+R_4}+\sqrt{C_1}\sqrt{C_3}\sqrt{R_3}\sqrt{R_4}\sqrt{\frac{g_m}{R_1g_m+1}}\sqrt{2R_3+R_4}}{2C_1R_1R_3g_m+C_1R_1R_4g_m+2C_1R_3+C_1R_4+C_3R_3R_4g_m}$$

$$\text{wo: } \sqrt{\frac{2R_3g_m+R_4g_m}{C_1C_3R_1R_3R_4g_m+C_1C_3R_3R_4}}$$

$$\text{bandwidth: } \frac{\sqrt{\frac{2R_3g_m+R_4g_m}{C_1C_3R_1R_3R_4g_m+C_1C_3R_3R_4}}(2C_1R_1R_3g_m+C_1R_1R_4g_m+2C_1R_3+C_1R_4+C_3R_3R_4g_m)}{\sqrt{C_1}\sqrt{C_3}R_1\sqrt{R_3}\sqrt{R_4g_m}\sqrt{\frac{g_m}{R_1g_m+1}}\sqrt{2R_3+R_4}+\sqrt{C_1}\sqrt{C_3}\sqrt{R_3}\sqrt{R_4}\sqrt{\frac{g_m}{R_1g_m+1}}\sqrt{2R_3+R_4}}$$

$$\text{K-LP: } \frac{R_3R_4}{2R_3+R_4}$$

$$\text{K-HP: } 0$$

$$\text{K-BP: } \frac{C_1R_1R_3R_4g_m}{2C_1R_1R_3g_m+C_1R_1R_4g_m+2C_1R_3+C_1R_4+C_3R_3R_4g_m}$$

$$\text{Qz: None}$$

$$\text{Wz: None}$$

8.12 X-INVALID-NUMER-12 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \frac{R_3}{C_3 R_3 s + 1}, \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_1 R_1 R_3 g_m s + R_3 g_m}{g_m + s^2 (C_1 C_3 R_1 R_3 g_m + C_1 C_3 R_3 + 2C_1 C_4 R_1 R_3 g_m + 2C_1 C_4 R_3) + s (C_1 R_1 g_m + C_1 + C_3 R_3 g_m + 2C_4 R_3 g_m)}$$

Parameters:

$$\text{Q: } \frac{\sqrt{C_1}C_3R_1\sqrt{R_3g_m}\sqrt{\frac{g_m}{C_3R_1g_m+C_3+2C_4R_1g_m+2C_4}}+\sqrt{C_1}C_3\sqrt{R_3}\sqrt{\frac{g_m}{C_3R_1g_m+C_3+2C_4R_1g_m+2C_4}}+2\sqrt{C_1}C_4R_1\sqrt{R_3g_m}\sqrt{\frac{g_m}{C_3R_1g_m+C_3+2C_4R_1g_m+2C_4}}+2\sqrt{C_1}C_4\sqrt{R_3}\sqrt{\frac{g_m}{C_3R_1g_m+C_3+2C_4R_1g_m+2C_4}}}{C_1R_1g_m+C_1+C_3R_3g_m+2C_4R_3g_m}$$

$$\text{wo: } \sqrt{\frac{g_m}{C_1C_3R_1R_3g_m+C_1C_3R_3+2C_1C_4R_1R_3g_m+2C_1C_4R_3}}$$

$$\text{bandwidth: } \frac{\sqrt{\frac{g_m}{C_1C_3R_1R_3g_m+C_1C_3R_3+2C_1C_4R_1R_3g_m+2C_1C_4R_3}}(C_1R_1g_m+C_1+C_3R_3g_m+2C_4R_3g_m)}{\sqrt{C_1}C_3R_1\sqrt{R_3g_m}\sqrt{\frac{g_m}{C_3R_1g_m+C_3+2C_4R_1g_m+2C_4}}+\sqrt{C_1}C_3\sqrt{R_3}\sqrt{\frac{g_m}{C_3R_1g_m+C_3+2C_4R_1g_m+2C_4}}+2\sqrt{C_1}C_4R_1\sqrt{R_3g_m}\sqrt{\frac{g_m}{C_3R_1g_m+C_3+2C_4R_1g_m+2C_4}}+2\sqrt{C_1}C_4\sqrt{R_3}\sqrt{\frac{g_m}{C_3R_1g_m+C_3+2C_4R_1g_m+2C_4}}}$$

$$\text{K-LP: } R_3$$

$$\text{K-HP: } 0$$

$$\text{K-BP: } \frac{C_1R_1R_3g_m}{C_1R_1g_m+C_1+C_3R_3g_m+2C_4R_3g_m}$$

$$\text{Qz: None}$$

$$\text{Wz: None}$$

8.13 X-INVALID-NUMER-13 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \frac{R_3}{C_3 R_3 s + 1}, \frac{R_4}{C_4 R_4 s + 1}, \infty, \infty \right)$

$$H(s) = \frac{C_1 R_1 R_3 R_4 g_m s + R_3 R_4 g_m}{2 R_3 g_m + R_4 g_m + s^2 (C_1 C_3 R_1 R_3 R_4 g_m + C_1 C_3 R_3 R_4 + 2 C_1 C_4 R_1 R_3 R_4 g_m + 2 C_1 C_4 R_3 R_4) + s (2 C_1 R_1 R_3 g_m + C_1 R_1 R_4 g_m + 2 C_1 R_3 + C_1 R_4 + C_3 R_3 R_4 g_m + 2 C_4 R_3 R_4 g_m)}$$

Parameters:

Q: $\frac{\sqrt{C_1 C_3 R_1} \sqrt{R_3} \sqrt{R_4 g_m} \sqrt{\frac{g_m}{C_3 R_1 g_m + C_3 + 2 C_4 R_1 g_m + 2 C_4}} \sqrt{2 R_3 + R_4} + \sqrt{C_1} C_3 \sqrt{R_3} \sqrt{R_4} \sqrt{\frac{g_m}{C_3 R_1 g_m + C_3 + 2 C_4 R_1 g_m + 2 C_4}} \sqrt{2 R_3 + R_4} + 2 \sqrt{C_1} C_4 R_1 \sqrt{R_3} \sqrt{R_4} g_m \sqrt{\frac{g_m}{C_3 R_1 g_m + C_3 + 2 C_4 R_1 g_m + 2 C_4}} \sqrt{2 R_3 + R_4} + 2 \sqrt{C_1} C_4 \sqrt{R_3} \sqrt{R_4} \sqrt{\frac{g_m}{C_3 R_1 g_m + C_3 + 2 C_4 R_1 g_m + 2 C_4}} \sqrt{2 R_3 + R_4}}{2 C_1 R_1 R_3 g_m + C_1 R_1 R_4 g_m + 2 C_1 R_3 + C_1 R_4 + C_3 R_3 R_4 g_m + 2 C_4 R_3 R_4 g_m}$

wo: $\sqrt{\frac{2 R_3 g_m + R_4 g_m}{C_1 C_3 R_1 R_3 R_4 g_m + C_1 C_3 R_3 R_4 + 2 C_1 C_4 R_1 R_3 R_4 g_m + 2 C_1 C_4 R_3 R_4}}$

bandwidth: $\frac{\sqrt{C_1 C_3 R_1} \sqrt{R_3} \sqrt{R_4 g_m} \sqrt{\frac{g_m}{C_3 R_1 g_m + C_3 + 2 C_4 R_1 g_m + 2 C_4}} \sqrt{2 R_3 + R_4} + \sqrt{C_1} C_3 \sqrt{R_3} \sqrt{R_4} \sqrt{\frac{g_m}{C_3 R_1 g_m + C_3 + 2 C_4 R_1 g_m + 2 C_4}} \sqrt{2 R_3 + R_4} + 2 \sqrt{C_1} C_4 R_1 \sqrt{R_3} \sqrt{R_4} g_m \sqrt{\frac{g_m}{C_3 R_1 g_m + C_3 + 2 C_4 R_1 g_m + 2 C_4}} \sqrt{2 R_3 + R_4} + 2 \sqrt{C_1} C_4 \sqrt{R_3} \sqrt{R_4} \sqrt{\frac{g_m}{C_3 R_1 g_m + C_3 + 2 C_4 R_1 g_m + 2 C_4}} \sqrt{2 R_3 + R_4}}{\sqrt{\frac{2 R_3 g_m + R_4 g_m}{C_1 C_3 R_1 R_3 R_4 g_m + C_1 C_3 R_3 R_4 + 2 C_1 C_4 R_1 R_3 R_4 g_m + 2 C_1 C_4 R_3 R_4}} (2 C_1 R_1 R_3 g_m + C_1 R_1 R_4 g_m + 2 C_1 R_3 + C_1 R_4 + C_3 R_3 R_4 g_m + 2 C_4 R_3 R_4 g_m)}$

K-LP: $\frac{R_3 R_4}{2 R_3 + R_4}$

K-HP: 0

K-BP: $\frac{C_1 R_1 R_3 R_4 g_m}{2 C_1 R_1 R_3 g_m + C_1 R_1 R_4 g_m + 2 C_1 R_3 + C_1 R_4 + C_3 R_3 R_4 g_m + 2 C_4 R_3 R_4 g_m}$

Qz: None

Wz: None

9 X-INVALID-ORDER

9.1 X-INVALID-ORDER-1 $Z(s) = (R_1, \infty, R_3, R_4, \infty, \infty)$

$$H(s) = \frac{R_1 R_3 R_4 g_m}{2 R_1 R_3 g_m + R_1 R_4 g_m + 2 R_3 + R_4}$$

9.2 X-INVALID-ORDER-2 $Z(s) = \left(R_1, \infty, R_3, \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{R_1 R_3 g_m}{R_1 g_m + s (2 C_4 R_1 R_3 g_m + 2 C_4 R_3) + 1}$$

9.3 X-INVALID-ORDER-3 $Z(s) = \left(R_1, \infty, R_3, \frac{R_4}{C_4 R_4 s + 1}, \infty, \infty \right)$

$$H(s) = \frac{R_1 R_3 R_4 g_m}{2 R_1 R_3 g_m + R_1 R_4 g_m + 2 R_3 + R_4 + s (2 C_4 R_1 R_3 R_4 g_m + 2 C_4 R_3 R_4)}$$

9.4 X-INVALID-ORDER-4 $Z(s) = \left(R_1, \infty, R_3, R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_4 R_1 R_3 R_4 g_m s + R_1 R_3 g_m}{R_1 g_m + s (2 C_4 R_1 R_3 g_m + C_4 R_1 R_4 g_m + 2 C_4 R_3 + C_4 R_4) + 1}$$

9.5 X-INVALID-ORDER-5 $Z(s) = \left(R_1, \infty, \frac{1}{C_3 s}, R_4, \infty, \infty \right)$

$$H(s) = \frac{R_1 R_4 g_m}{2 R_1 g_m + s (C_3 R_1 R_4 g_m + C_3 R_4) + 2}$$

9.6 X-INVALID-ORDER-6 $Z(s) = \left(R_1, \infty, \frac{1}{C_3 s}, \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{R_1 g_m}{s (C_3 R_1 g_m + C_3 + 2 C_4 R_1 g_m + 2 C_4)}$$

$$\mathbf{9.7 \quad X-INVALID-ORDER-7} \quad Z(s) = \left(R_1, \infty, \frac{1}{C_3 s}, \frac{R_4}{C_4 R_4 s + 1}, \infty, \infty \right)$$

$$H(s) = \frac{R_1 R_4 g_m}{2R_1 g_m + s(C_3 R_1 R_4 g_m + C_3 R_4 + 2C_4 R_1 R_4 g_m + 2C_4 R_4) + 2}$$

$$\mathbf{9.8 \quad X-INVALID-ORDER-8} \quad Z(s) = \left(R_1, \infty, \frac{1}{C_3 s}, R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$$

$$H(s) = \frac{C_4 R_1 R_4 g_m s + R_1 g_m}{s^2(C_3 C_4 R_1 R_4 g_m + C_3 C_4 R_4) + s(C_3 R_1 g_m + C_3 + 2C_4 R_1 g_m + 2C_4)}$$

$$\mathbf{9.9 \quad X-INVALID-ORDER-9} \quad Z(s) = \left(R_1, \infty, \frac{R_3}{C_3 R_3 s + 1}, R_4, \infty, \infty \right)$$

$$H(s) = \frac{R_1 R_3 R_4 g_m}{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4 + s(C_3 R_1 R_3 R_4 g_m + C_3 R_3 R_4)}$$

$$\mathbf{9.10 \quad X-INVALID-ORDER-10} \quad Z(s) = \left(R_1, \infty, \frac{R_3}{C_3 R_3 s + 1}, \frac{1}{C_4 s}, \infty, \infty \right)$$

$$H(s) = \frac{R_1 R_3 g_m}{R_1 g_m + s(C_3 R_1 R_3 g_m + C_3 R_3 + 2C_4 R_1 R_3 g_m + 2C_4 R_3) + 1}$$

$$\mathbf{9.11 \quad X-INVALID-ORDER-11} \quad Z(s) = \left(R_1, \infty, \frac{R_3}{C_3 R_3 s + 1}, \frac{R_4}{C_4 R_4 s + 1}, \infty, \infty \right)$$

$$H(s) = \frac{R_1 R_3 R_4 g_m}{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4 + s(C_3 R_1 R_3 R_4 g_m + C_3 R_3 R_4 + 2C_4 R_1 R_3 R_4 g_m + 2C_4 R_3 R_4)}$$

$$\mathbf{9.12 \quad X-INVALID-ORDER-12} \quad Z(s) = \left(R_1, \infty, R_3 + \frac{1}{C_3 s}, R_4, \infty, \infty \right)$$

$$H(s) = \frac{C_3 R_1 R_3 R_4 g_m s + R_1 R_4 g_m}{2R_1 g_m + s(2C_3 R_1 R_3 g_m + C_3 R_1 R_4 g_m + 2C_3 R_3 + C_3 R_4) + 2}$$

$$\mathbf{9.13 \quad X-INVALID-ORDER-13} \quad Z(s) = \left(R_1, \infty, R_3 + \frac{1}{C_3 s}, \frac{1}{C_4 s}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 R_1 R_3 g_m s + R_1 g_m}{s^2(2C_3 C_4 R_1 R_3 g_m + 2C_3 C_4 R_3) + s(C_3 R_1 g_m + C_3 + 2C_4 R_1 g_m + 2C_4)}$$

$$\mathbf{9.14 \quad X-INVALID-ORDER-14} \quad Z(s) = \left(R_1, \infty, R_3 + \frac{1}{C_3 s}, R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 C_4 R_1 R_3 R_4 g_m s^2 + R_1 g_m + s(C_3 R_1 R_3 g_m + C_4 R_1 R_4 g_m)}{s^2(2C_3 C_4 R_1 R_3 g_m + C_3 C_4 R_1 R_4 g_m + 2C_3 C_4 R_3 + C_3 C_4 R_4) + s(C_3 R_1 g_m + C_3 + 2C_4 R_1 g_m + 2C_4)}$$

$$\mathbf{9.15 \quad X-INVALID-ORDER-15} \quad Z(s) = \left(\frac{1}{C_1 s}, \infty, R_3, R_4, \infty, \infty \right)$$

$$H(s) = \frac{R_3 R_4 g_m}{2R_3 g_m + R_4 g_m + s(2C_1 R_3 + C_1 R_4)}$$

$$\mathbf{9.16 \quad X-INVALID-ORDER-16} \quad Z(s) = \left(\frac{1}{C_1 s}, \infty, \frac{1}{C_3 s}, \frac{1}{C_4 s}, \infty, \infty \right)$$

$$H(s) = \frac{g_m}{s^2(C_1 C_3 + 2C_1 C_4) + s(C_3 g_m + 2C_4 g_m)}$$

$$\mathbf{9.17 \quad X-INVALID-ORDER-17} \quad Z(s) = \left(\frac{1}{C_1 s}, \infty, \frac{1}{C_3 s}, R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$$

$$H(s) = \frac{C_4 R_4 g_m s + g_m}{C_1 C_3 C_4 R_4 s^3 + s^2 (C_1 C_3 + 2C_1 C_4 + C_3 C_4 R_4 g_m) + s (C_3 g_m + 2C_4 g_m)}$$

$$\mathbf{9.18 \quad X-INVALID-ORDER-18} \quad Z(s) = \left(\frac{1}{C_1 s}, \infty, \frac{R_3}{C_3 R_3 s + 1}, R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$$

$$H(s) = \frac{C_4 R_3 R_4 g_m s + R_3 g_m}{C_1 C_3 C_4 R_3 R_4 s^3 + g_m + s^2 (C_1 C_3 R_3 + 2C_1 C_4 R_3 + C_1 C_4 R_4 + C_3 C_4 R_3 R_4 g_m) + s (C_1 + C_3 R_3 g_m + 2C_4 R_3 g_m + C_4 R_4 g_m)}$$

$$\mathbf{9.19 \quad X-INVALID-ORDER-19} \quad Z(s) = \left(\frac{1}{C_1 s}, \infty, R_3 + \frac{1}{C_3 s}, \frac{1}{C_4 s}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 R_3 g_m s + g_m}{2C_1 C_3 C_4 R_3 s^3 + s^2 (C_1 C_3 + 2C_1 C_4 + 2C_3 C_4 R_3 g_m) + s (C_3 g_m + 2C_4 g_m)}$$

$$\mathbf{9.20 \quad X-INVALID-ORDER-20} \quad Z(s) = \left(\frac{1}{C_1 s}, \infty, R_3 + \frac{1}{C_3 s}, \frac{R_4}{C_4 R_4 s + 1}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 R_3 R_4 g_m s + R_4 g_m}{2C_1 C_3 C_4 R_3 R_4 s^3 + 2g_m + s^2 (2C_1 C_3 R_3 + C_1 C_3 R_4 + 2C_1 C_4 R_4 + 2C_3 C_4 R_3 R_4 g_m) + s (2C_1 + 2C_3 R_3 g_m + C_3 R_4 g_m + 2C_4 R_4 g_m)}$$

$$\mathbf{9.21 \quad X-INVALID-ORDER-21} \quad Z(s) = \left(\frac{1}{C_1 s}, \infty, R_3 + \frac{1}{C_3 s}, R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 C_4 R_3 R_4 g_m s^2 + g_m + s (C_3 R_3 g_m + C_4 R_4 g_m)}{s^3 (2C_1 C_3 C_4 R_3 + C_1 C_3 C_4 R_4) + s^2 (C_1 C_3 + 2C_1 C_4 + 2C_3 C_4 R_3 g_m + C_3 C_4 R_4 g_m) + s (C_3 g_m + 2C_4 g_m)}$$

$$\mathbf{9.22 \quad X-INVALID-ORDER-22} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, R_3, R_4, \infty, \infty \right)$$

$$H(s) = \frac{R_1 R_3 R_4 g_m}{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4 + s (2C_1 R_1 R_3 + C_1 R_1 R_4)}$$

$$\mathbf{9.23 \quad X-INVALID-ORDER-23} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \frac{1}{C_3 s}, \frac{1}{C_4 s}, \infty, \infty \right)$$

$$H(s) = \frac{R_1 g_m}{s^2 (C_1 C_3 R_1 + 2C_1 C_4 R_1) + s (C_3 R_1 g_m + C_3 + 2C_4 R_1 g_m + 2C_4)}$$

$$\mathbf{9.24 \quad X-INVALID-ORDER-24} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \frac{1}{C_3 s}, R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$$

$$H(s) = \frac{C_4 R_1 R_4 g_m s + R_1 g_m}{C_1 C_3 C_4 R_1 R_4 s^3 + s^2 (C_1 C_3 R_1 + 2C_1 C_4 R_1 + C_3 C_4 R_1 R_4 g_m + C_3 C_4 R_4) + s (C_3 R_1 g_m + C_3 + 2C_4 R_1 g_m + 2C_4)}$$

$$\mathbf{9.25 \quad X-INVALID-ORDER-25} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \frac{R_3}{C_3 R_3 s + 1}, R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$$

$$H(s) = \frac{C_4 R_1 R_3 R_4 g_m s + R_1 R_3 g_m}{C_1 C_3 C_4 R_1 R_3 R_4 s^3 + R_1 g_m + s^2 (C_1 C_3 R_1 R_3 + 2C_1 C_4 R_1 R_3 + C_1 C_4 R_1 R_4 + C_3 C_4 R_1 R_3 R_4 g_m + C_3 C_4 R_3 R_4) + s (C_1 R_1 + C_3 R_1 R_3 g_m + C_3 R_3 + 2C_4 R_1 R_3 g_m + C_4 R_1 R_4 g_m + 2C_4 R_3 + C_4 R_4) + 1}$$

$$\mathbf{9.26 \quad X-INVALID-ORDER-26} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, R_3 + \frac{1}{C_3 s}, \frac{1}{C_4 s}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 R_1 R_3 g_m s + R_1 g_m}{2C_1 C_3 C_4 R_1 R_3 s^3 + s^2 (C_1 C_3 R_1 + 2C_1 C_4 R_1 + 2C_3 C_4 R_1 R_3 g_m + 2C_3 C_4 R_3) + s (C_3 R_1 g_m + C_3 + 2C_4 R_1 g_m + 2C_4)}$$

9.27 X-INVALID-ORDER-27 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, R_3 + \frac{1}{C_3 s}, \frac{R_4}{C_4 R_4 s + 1}, \infty, \infty \right)$

$$H(s) = \frac{C_3 R_1 R_3 R_4 g_m s + R_1 R_4 g_m}{2C_1 C_3 C_4 R_1 R_3 R_4 s^3 + 2R_1 g_m + s^2 (2C_1 C_3 R_1 R_3 + C_1 C_3 R_1 R_4 + 2C_1 C_4 R_1 R_4 + 2C_3 C_4 R_1 R_3 R_4 g_m + 2C_3 C_4 R_3 R_4) + s (2C_1 R_1 + 2C_3 R_1 R_3 g_m + C_3 R_1 R_4 g_m + 2C_3 R_3 + C_3 R_4 + 2C_4 R_1 R_4 g_m + 2C_4 R_4) + 2}$$

9.28 X-INVALID-ORDER-28 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, R_3 + \frac{1}{C_3 s}, R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_3 C_4 R_1 R_3 R_4 g_m s^2 + R_1 g_m + s (C_3 R_1 R_3 g_m + C_4 R_1 R_4 g_m)}{s^3 (2C_1 C_3 C_4 R_1 R_3 + C_1 C_3 C_4 R_1 R_4) + s^2 (C_1 C_3 R_1 + 2C_1 C_4 R_1 + 2C_3 C_4 R_1 R_3 g_m + C_3 C_4 R_1 R_4 g_m + 2C_3 C_4 R_3 + C_3 C_4 R_4) + s (C_3 R_1 g_m + C_3 + 2C_4 R_1 g_m + 2C_4)}$$

9.29 X-INVALID-ORDER-29 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, R_3, R_4, \infty, \infty \right)$

$$H(s) = \frac{C_1 R_1 R_3 R_4 g_m s + R_3 R_4 g_m}{2R_3 g_m + R_4 g_m + s (2C_1 R_1 R_3 g_m + C_1 R_1 R_4 g_m + 2C_1 R_3 + C_1 R_4)}$$

9.30 X-INVALID-ORDER-30 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \frac{1}{C_3 s}, \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_1 R_1 g_m s + g_m}{s^2 (C_1 C_3 R_1 g_m + C_1 C_3 + 2C_1 C_4 R_1 g_m + 2C_1 C_4) + s (C_3 g_m + 2C_4 g_m)}$$

9.31 X-INVALID-ORDER-31 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \frac{1}{C_3 s}, R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_4 R_1 R_4 g_m s^2 + g_m + s (C_1 R_1 g_m + C_4 R_4 g_m)}{s^3 (C_1 C_3 C_4 R_1 R_4 g_m + C_1 C_3 C_4 R_4) + s^2 (C_1 C_3 R_1 g_m + C_1 C_3 + 2C_1 C_4 R_1 g_m + 2C_1 C_4 + C_3 C_4 R_4 g_m) + s (C_3 g_m + 2C_4 g_m)}$$

9.32 X-INVALID-ORDER-32 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \frac{R_3}{C_3 R_3 s + 1}, R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_4 R_1 R_3 R_4 g_m s^2 + R_3 g_m + s (C_1 R_1 R_3 g_m + C_4 R_3 R_4 g_m)}{g_m + s^3 (C_1 C_3 C_4 R_1 R_3 R_4 g_m + C_1 C_3 C_4 R_3 R_4) + s^2 (C_1 C_3 R_1 R_3 g_m + C_1 C_3 R_3 + 2C_1 C_4 R_1 R_3 g_m + C_1 C_4 R_1 R_4 g_m + 2C_1 C_4 R_3 + C_1 C_4 R_4 + C_3 C_4 R_3 R_4 g_m) + s (C_1 R_1 g_m + C_1 + C_3 R_3 g_m + 2C_4 R_3 g_m + C_4 R_4 g_m)}$$

9.33 X-INVALID-ORDER-33 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, R_3 + \frac{1}{C_3 s}, \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_3 R_1 R_3 g_m s^2 + g_m + s (C_1 R_1 g_m + C_3 R_3 g_m)}{s^3 (2C_1 C_3 C_4 R_1 R_3 g_m + 2C_1 C_3 C_4 R_3) + s^2 (C_1 C_3 R_1 g_m + C_1 C_3 + 2C_1 C_4 R_1 g_m + 2C_1 C_4 + 2C_3 C_4 R_3 g_m) + s (C_3 g_m + 2C_4 g_m)}$$

9.34 X-INVALID-ORDER-34 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, R_3 + \frac{1}{C_3 s}, \frac{R_4}{C_4 R_4 s + 1}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_3 R_1 R_3 R_4 g_m s^2 + R_4 g_m + s (C_1 R_1 R_4 g_m + C_3 R_3 R_4 g_m)}{2g_m + s^3 (2C_1 C_3 C_4 R_1 R_3 R_4 g_m + 2C_1 C_3 C_4 R_3 R_4) + s^2 (2C_1 C_3 R_1 R_3 g_m + C_1 C_3 R_1 R_4 g_m + 2C_1 C_3 R_3 + C_1 C_3 R_4 + 2C_1 C_4 R_1 R_4 g_m + 2C_1 C_4 R_4 + 2C_3 C_4 R_3 R_4 g_m) + s (2C_1 R_1 g_m + 2C_1 + 2C_3 R_3 g_m + C_3 R_4 g_m + 2C_4 R_4 g_m)}$$

9.35 X-INVALID-ORDER-35 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, R_3 + \frac{1}{C_3 s}, R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_3 C_4 R_1 R_3 R_4 g_m s^3 + g_m + s^2 (C_1 C_3 R_1 R_3 g_m + C_1 C_4 R_1 R_4 g_m + C_3 C_4 R_3 R_4 g_m) + s (C_1 R_1 g_m + C_3 R_3 g_m + C_4 R_4 g_m)}{s^3 (2C_1 C_3 C_4 R_1 R_3 g_m + C_1 C_3 C_4 R_1 R_4 g_m + 2C_1 C_3 C_4 R_3 + C_1 C_3 C_4 R_4) + s^2 (C_1 C_3 R_1 g_m + C_1 C_3 + 2C_1 C_4 R_1 g_m + 2C_1 C_4 + 2C_3 C_4 R_3 g_m + C_3 C_4 R_4 g_m) + s (C_3 g_m + 2C_4 g_m)}$$

10 X-INVALID-WZ

10.1 X-INVALID-WZ-1 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, R_3, R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_4 R_1 R_3 R_4 g_m s^2 + R_3 g_m + s (C_1 R_1 R_3 g_m + C_4 R_3 R_4 g_m)}{g_m + s^2 (2C_1 C_4 R_1 R_3 g_m + C_1 C_4 R_1 R_4 g_m + 2C_1 C_4 R_3 + C_1 C_4 R_4) + s (C_1 R_1 g_m + C_1 + 2C_4 R_3 g_m + C_4 R_4 g_m)}$$

Parameters:

$$\text{Q: } \frac{2\sqrt{C_1}\sqrt{C_4}R_1R_3g_m\sqrt{\frac{g_m}{2R_1R_3g_m+R_1R_4g_m+2R_3+R_4}}+\sqrt{C_1}\sqrt{C_4}R_1R_4g_m\sqrt{\frac{g_m}{2R_1R_3g_m+R_1R_4g_m+2R_3+R_4}}+2\sqrt{C_1}\sqrt{C_4}R_3\sqrt{\frac{g_m}{2R_1R_3g_m+R_1R_4g_m+2R_3+R_4}}+\sqrt{C_1}\sqrt{C_4}R_4\sqrt{\frac{g_m}{2R_1R_3g_m+R_1R_4g_m+2R_3+R_4}}}{C_1R_1g_m+C_1+2C_4R_3g_m+C_4R_4g_m}$$

$$\text{wo: } \sqrt{\frac{g_m}{2C_1C_4R_1R_3g_m+C_1C_4R_1R_4g_m+2C_1C_4R_3+C_1C_4R_4}}$$

$$\text{bandwidth: } \frac{\sqrt{\frac{g_m}{2C_1C_4R_1R_3g_m+C_1C_4R_1R_4g_m+2C_1C_4R_3+C_1C_4R_4}}(C_1R_1g_m+C_1+2C_4R_3g_m+C_4R_4g_m)}{2\sqrt{C_1}\sqrt{C_4}R_1R_3g_m\sqrt{\frac{g_m}{2R_1R_3g_m+R_1R_4g_m+2R_3+R_4}}+\sqrt{C_1}\sqrt{C_4}R_1R_4g_m\sqrt{\frac{g_m}{2R_1R_3g_m+R_1R_4g_m+2R_3+R_4}}+2\sqrt{C_1}\sqrt{C_4}R_3\sqrt{\frac{g_m}{2R_1R_3g_m+R_1R_4g_m+2R_3+R_4}}+\sqrt{C_1}\sqrt{C_4}R_4\sqrt{\frac{g_m}{2R_1R_3g_m+R_1R_4g_m+2R_3+R_4}}}$$

K-LP: R_3

$$\text{K-HP: } \frac{R_1R_3R_4g_m}{2R_1R_3g_m+R_1R_4g_m+2R_3+R_4}$$

$$\text{K-BP: } \frac{C_1R_1R_3g_m+C_4R_3R_4g_m}{C_1R_1g_m+C_1+2C_4R_3g_m+C_4R_4g_m}$$

Qz: None

$$\text{Wz: } \frac{1}{\sqrt{C_1}\sqrt{C_4}\sqrt{R_1}\sqrt{R_4}}$$

10.2 X-INVALID-WZ-2 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, R_3 + \frac{1}{C_3 s}, R_4, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_3 R_1 R_3 R_4 g_m s^2 + R_4 g_m + s (C_1 R_1 R_4 g_m + C_3 R_3 R_4 g_m)}{2g_m + s^2 (2C_1 C_3 R_1 R_3 g_m + C_1 C_3 R_1 R_4 g_m + 2C_1 C_3 R_3 + C_1 C_3 R_4) + s (2C_1 R_1 g_m + 2C_1 + 2C_3 R_3 g_m + C_3 R_4 g_m)}$$

Parameters:

$$\text{Q: } \frac{2\sqrt{2}\sqrt{C_1}\sqrt{C_3}R_1R_3g_m\sqrt{\frac{g_m}{2R_1R_3g_m+R_1R_4g_m+2R_3+R_4}}+\sqrt{2}\sqrt{C_1}\sqrt{C_3}R_1R_4g_m\sqrt{\frac{g_m}{2R_1R_3g_m+R_1R_4g_m+2R_3+R_4}}+2\sqrt{2}\sqrt{C_1}\sqrt{C_3}R_3\sqrt{\frac{g_m}{2R_1R_3g_m+R_1R_4g_m+2R_3+R_4}}+\sqrt{2}\sqrt{C_1}\sqrt{C_3}R_4\sqrt{\frac{g_m}{2R_1R_3g_m+R_1R_4g_m+2R_3+R_4}}}{2C_1R_1g_m+2C_1+2C_3R_3g_m+C_3R_4g_m}$$

$$\text{wo: } \sqrt{2}\sqrt{\frac{g_m}{2C_1C_3R_1R_3g_m+C_1C_3R_1R_4g_m+2C_1C_3R_3+C_1C_3R_4}}$$

$$\text{bandwidth: } \frac{\sqrt{2}\sqrt{\frac{g_m}{2C_1C_3R_1R_3g_m+C_1C_3R_1R_4g_m+2C_1C_3R_3+C_1C_3R_4}}(2C_1R_1g_m+2C_1+2C_3R_3g_m+C_3R_4g_m)}{2\sqrt{2}\sqrt{C_1}\sqrt{C_3}R_1R_3g_m\sqrt{\frac{g_m}{2R_1R_3g_m+R_1R_4g_m+2R_3+R_4}}+\sqrt{2}\sqrt{C_1}\sqrt{C_3}R_1R_4g_m\sqrt{\frac{g_m}{2R_1R_3g_m+R_1R_4g_m+2R_3+R_4}}+2\sqrt{2}\sqrt{C_1}\sqrt{C_3}R_3\sqrt{\frac{g_m}{2R_1R_3g_m+R_1R_4g_m+2R_3+R_4}}+\sqrt{2}\sqrt{C_1}\sqrt{C_3}R_4\sqrt{\frac{g_m}{2R_1R_3g_m+R_1R_4g_m+2R_3+R_4}}}$$

K-LP: $\frac{R_4}{2}$

$$\text{K-HP: } \frac{R_1R_3R_4g_m}{2R_1R_3g_m+R_1R_4g_m+2R_3+R_4}$$

$$\text{K-BP: } \frac{C_1R_1R_4g_m+C_3R_3R_4g_m}{2C_1R_1g_m+2C_1+2C_3R_3g_m+C_3R_4g_m}$$

Qz: None

$$\text{Wz: } \frac{1}{\sqrt{C_1}\sqrt{C_3}\sqrt{R_1}\sqrt{R_3}}$$

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